B.Tech. (Computer Engg.) V Semester Examinations
Session 2026-27

Time: 1 hr. **Computer Networks (CSC - 503) Sessional-I** Max.Marks: 12

Note: Attempt All Questions

1. CO1	Consider the network: Host A → Router R1 → Router R2 → Host B. Host A has IP 192.168.1.10, MAC AA, and uses source port 5000. Host B has IP 172.16.1.10, MAC BB, and uses destination port 80. Explain how a packet travels from Host A to Host B through R1 and R2. For each link (A→R1, R1→R2, R2→B), specify the source/destination MAC addresses, source/destination IP addresses, and source/destination port numbers.	3	(BL3)
2. CO1	Suppose two computers communicate using a Virtual Circuit network through three intermediate switches. Explain how the switches maintain information about the established Virtual Circuit. Show the scenario how VCIs help the network to identify and forward packets belonging to different connections.	3	(BL2,BL
3. CO4	Consider a network in which multiple internal devices use private IP addresses and access the Internet through a Network Address Translation (NAT) device. Explain how the NAT device enables two different internal devices to communicate with the same external server simultaneously. Illustrate the scenario by taking suitable example.	3	(BL2,BL
4. CO4	A company is assigned 100.168.16.175/26 network for its internal use. Divide this network into 3 subnets of size 8, 16 and 8. Find the subnet ID bits, broadcast addresses, range of valid host IP addresses.	3	(BL5)